Tomojit Ghosh

ASSISTANT PROFESSOR · DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Education

Colorado State University *Fort Collins, CO, USA*
DOCTORATE OF PHILOSOPHY IN COMPUTER SCIENCE *2017 - 2022*
- Thesis: Convex and Non-Convex Optimization using Centroid-Encoding for Visualization, Classification, and Feature Selection
- Advisor: Dr. Michael Kirby

Colorado State University *Fort Collins, CO, USA*
MASTER OF SCIENCE IN COMPUTER SCIENCE *2013 - 2017*
- Thesis: Supervised and unsupervised training of deep autoencoder
- Advisor: Dr. Charles Anderson

Indian Institute of Engineering Science and Technology, Shibpur (IIEST) *Howrah,WB, India*
BACHELOR OF ENGINEERING, INFORMATION TECHNOLOGY *2003 - 2006*

Professional Experience

08/2024 : current	**Assistant Professor**, Department of Computer Science and Eng., Wright State University
08/2023 : 07/2024	**Assistant Professor**, Department of Computer Science and Eng., University of Tennessee at Chattanooga
08/2022 : 06/2023	**Post Doctoral Fellow**, Department of Mathematics, Colorado State University
06/2014 : 08/2014	**Summer Internship**, Colorado Neurological Institute, US
09/2008 : 06/2013	**Senior Software Developer**, Cognizant Technology Solutions, US
06/2006 : 08/2008	**Junior Software Developer**, Cognizant Technology Solutions, India

Research Interest

The aim of my research is to develop novel algorithms for visualization, feature selection, and classification using the underlying geometric structure of data. I'm generally interested in the theory and application of machine learning, artificial intelligence, optimization, modeling and simulation, and the application of pattern recognition and machine learning in Omics (transcriptomic, genomics, proteomics, and metabolomics) data.

Publications

PUBLISHED

15. Linear centroid encoder for supervised principal component analysis
Tomojit Ghosh, and Michael Kirby.
journal =Pattern Recognition, year=2024, volume=155, publisher=Elsevier.

14. A Multi-domain Multi-task Approach for Feature Selection from Bulk RNA Datasets
Karim Salta, Tomojit Ghosh and Michael Kirby .

24^{th} International Conference on Computational Science (ICCS), Malaga, Spain, pages=31–45, organization=Springer, year=2024.

13. Enhanced UAV Security: Optimizing Accuracy and Efficiency with MSBCE Feature Selection
Tomojit Ghosh and Shahnewaz Karim Sakib .
12^{th} International Symposium on Digital Forensics and Security (to be appeared), year=2024.

12. Evaluation of Privacy-Utility Tradeoff in Generative Adversarial Network Variants
Shahnewaz Karim Sakib, and Tomojit Ghosh.
12^{th} International Symposium on Digital Forensics and Security (to be appeared), year=2024.

11. Feature Selection on Big Data using Masked Sparse Bottleneck Centroid-Encoder
Tomojit Ghosh, and Michael Kirby.
IEEE International Conference on Big Data (Acceptance rate = 17.49%), year=2023.

10. Sparse Linear Centroid-Encoder: A Biomarker Selection tool for High Dimensional Biological Data
Tomojit Ghosh, Karim Karimov, and Michael Kirby.
IEEE BIBM Conference proceedings (Workshop: ML & AI in Bioinformatics and Medical Informatics - MABM), year=2023.

9. Nonlinear Feature Selection using Sparsity-promoted Centroid-Encoder
Tomojit Ghosh, and Michael Kirby.
Neural Computing and Applications- NCAA (Springer Publishing Group), year=2023, volume=35, doi=10.1007/s00521-023-08938-7,url=https://doi.org/10.1007/s00521-023-08938-7.

8. Using Machine Learning to Determine the Time of Exposure to Infection by a Respiratory Pathogen
Kartikay Sharma, Manuchehr Aminian, Tomojit Ghosh, Xiaoyu Liu, and Michael Kirby.
Scientific reports (Nature Publishing Group), 2023.

7. Supervised Dimensionality Reduction and Visualization using Centroid-Encoder
Tomojit Ghosh and Michael Kirby.
Journal of Machine Learning Research (JMLR), year=2022, volume=23, url=http://jmlr.org/papers/v23/20-188.html.

6. Early prognosis of respiratory virus shedding in humans
Manuchehr Aminian, Tomojit Ghosh, Amy Peterson, Angela L Rasmussen, Shannon Stiverson, Kartikay Sharma, and Michael Kirby.
Scientific reports (Nature Publishing Group), 2021.

5. New Tools for the Visualization of Biological Pathways
Tomojit Ghosh, Xiaofeng Ma, Michael Kirby.
Methods (Elsevier), pages=26–33, 2018.

4. A sequential simplex algorithm for automatic data and center selecting radial basis functions
Xiaofeng Ma, Tomojit Ghosh, Michael Kirby .
International Joint Conference on Neural Networks (IJCNN), pages=549–556, 2017.

3. Dantzig-Selector Radial Basis Function Learning with Nonconvex Refinement
Tomojit Ghosh, Xiaofeng Ma, Michael Kirby .
Advances in Time Series Analysis and Forecasting: Selected Contributions from ITISE 2016, Springer International Publishing 2017, pages=313–327, 2017.

2. Explorations in Very Early Prognosis of the Human Immune Response to Influenza
Mmanu Chaturvedi, Tomojit Ghosh, Michael Kirby, Xiaoyu Liu, Xiaofeng Ma, Shannon Stiverson.
Proceedings of the 7th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics, pages=562–570, 2016.

1. Sparse skew radial basis functions for time-series prediction
Tomojit Ghosh, Michael Kirby and Xiaofeng Ma.
Proceedings International Work Conference on Time Series Analysis, pages=296–307, 2016.

In Review

1. **A Convex formulation for linear discriminant analysis**
2. **Generative AI for High-Dimensional Biological Data: A Comparative Study and Critical Review**

In Preparation

6. Isometric Centroid-Encoder

5. Review of Contrastive Learning

4. Review of Feature Selection with an Application to Biological data

3. An Algorithm for Supervised Multi Dimensional Scaling

2. Prototype-Encoder: An unsupervised model for automatic prototype selection, visualization with feature sparsity

1. Stochastic Centroid-Encoder: A nonlinear model for effective sample augmentation for biological data

Grant and Funding Activity

2025	**NSF CISE Core Proposal Submission (IIS)**, "Automated Interpretable Clustering for High-Dimensional, Low-Sample-Size Biological Data"(Request Amt. $391,001)	*NSF*
2025	**NIH NIBIB Proposal Submission**, "Artificial Intelligence for Respiratory Warning with Interpretable Sparse Embedding for Early Prediction of Pediatric Acute Respiratory Failure" (with Dayton Children's Hospital)(Request Amt. $478,806)	*NIH*
2025	**Amazon Research Award Proposal Submission**, "Sparse Energy-Aware AI Framework for Edge Device Sustainability"(Request Amt. $100,000)	*Amazon*
2024	**DoD Proposal Submission (AFRL Collaboration)**, "Development of Machine Learning Models for High-Altitude Hypoxic Response" (Requested amt: $76,716)	*DoD / AFRL*
2024	**NSF CRII Proposal Submission (IIS)**, "scPrototype-Encoder: Next-Generation Clustering for Automated Cell-Type and Marker Identification in scRNA-seq"(Requested amt: $174,182)	*NSF*

Teaching and Academic Employment

FA 2025	**CS4710/6710 (Introduction to Data Mining)**, Instructor	*WSU*
FA 2025	**CS4850/6850 (Foundation of AI)**, Instructor	*WSU*
SP 2025	**CS7900-02 (Algorithms for Biological Data)**, Instructor	*WSU*
FA 2024	**CS4850/6850 (Foundation of AI)**, Instructor	*WSU*
SP 2024	**CPSC5410 (Model Analysis and Simulation)**, Instructor	*UTC*
FA 2023	**CPSC1110 (Data Structures & Program Design with Java)**, Instructor	*UTC*
SP 2017 : SP 2022	**Research**, Graduate Research Assistant	*CSU*
FA 2016	**CS545 (Machine Learning)**, Graduate Teaching Assistant	*CSU*
SP 2016	**Research**, Graduate Research Assistant	*CSU*
FA 2015	**Research**, Graduate Research Assistant	*CSU*
SP 2015	**CS253 (Programming with C++)**, Graduate Teaching Assistant	*CSU*
FA 2014	**CS155/56/57 (Unix, C and Advanced C)**, Graduate Teaching Assistant	*CSU*
SP 2014	**CS253 (Programming with C++)**, Graduate Teaching Assistant	*CSU*
FA 2013	**CS155/56/57 (Unix, C and Advanced C)**, Graduate Teaching Assistant	*CSU*

Awards

April 22 2024	**Outstanding Tenured/Tenure-track Faculty Teaching Award**, College of Engineering and Computer Science, UTC	
April 22 2024	**Outstanding Faculty Teaching Award**, Department of Computer Science and Engineering, UTC	
April 18 2023	**Innovation in Natural Science**, CSU Demo Day STRATA	*$ 1,000*
2022	**Evolutionary Computing & Artificial Intelligence Fellowship**, Department of Computer Science, CSU	*$ 3,500*
2016	**ACM BigLS 2016 Student Travel award**, ACM BCB	*$ 1000*

Presentations/Talks

Presentations

5. Feature Selection on Big Data using Masked Sparse Bottleneck Centroid-Encoder. MCBIOS 2024, March 22-24, Emory University, Atlanta, GA.

4. Sparse Linear Centroid-Encoder: A Biomarker Selection tool for High Dimensional Biological Data. MCBIOS 2024, March 22-24, Emory University, Atlanta, GA.

3. Preliminary Analysis of Diagnostic Signatures in Transcriptomic Data for Bovine Respiratory Disease. Zoetis Journal Club Presentation, ZINC, August 2021, Online.

2. Centroid-encoder: A New Tool for Visualization, Classification and Feature Selection. CSU-Zoetis Journal Club Invitation, April 2020, Online.

1. Centroid-encoder Neural Networks for Data Visualization and Classification. The 3rd Annual Meeting of SIAM Central States Section, September 2017, CSU, Fort Collins, CO.

Research Talks

4. Analyzing Biological Data using Machine Learning. March 2023, University of Florida, Gainesville, FL. (Online)

3. Analyzing Biological Data using Machine Learning. Pritykin Lab, Lewis-Sigler Institute for Integrative Genomics, February 2023, Princeton University, Princeton, NJ.

2. Some Convex and Nonconvex Models to Analyze Biological Data Sets. Data Science Seminar, Department of Mathematics, November 2023, University of Tennessee, Knoxville, TN.

1. Centroid-encoder: A New Tool for Visualization, Classification and Feature Selection. Data Science Seminar, October 2019, CSU, Fort Collins, CO.

Poster Presentation

5. A Machine Learning Approach to Classifying Phoenix Sepsis in Pediatric Emergency Encounters: Phase One Results and Next Steps as Co-Author. 2025 AAP National Conference, Session: Council on Clinical Information Technology, Sept. 26-30, Denver, CO (to be held)

4. Sparse Linear Centroid-Encoder: A Convex Method for Feature Selection. Quantitative Cell and Molecular Biology Symposium 2023 (qCMB/UQ-Bio 2023), June 12 - June 13, Fort Collins, CO.

3. Biomarker Selection using Sparse Centroid-Encoder. CSU DemoDay 2023 (STRATA CSU 2023), April 18 , Fort Collins, CO.

2. Supervised Dimensionality Reduction and Visualization using Centroid-Encoder. Neural Information Processing Systems 2022 (NeurIPS 2022), November 28 - December 9, New Orleans, LA.

1. Biomarker Discovery via Nonlinear Function Approximation. ACM International Workshop on Big Data in Life Sciences (ACM BigLS), October 2016, Seattle, WA.

Mentoring

Nov 2025 - current	**Siva Rama Krishna Prasad Changala**, PhD Student	*WSU*
Aug 2025 - current	**Chikaosolu Amaechi**, PhD Student	*WSU*
Aug 2024 - current	**Sai Vijay Kumar Surineela**, Master's Thesis	*WSU*
Aug 2025 - current	**Venkata Thanoj Varma Atluri**, Master's Thesis	*WSU*
Aug 2025 - current	**Lauren Popp**, Undergraduate Research Student	*WSU*
Aug 2024 - Dec 2025	**Prathyusha Kanakamalla**, Master's Thesis	*WSU*
Aug 2024 - Dec 2025	**Harigovind Harikumar**, Master's Thesis	*WSU*
May 2024 - July 2024	**Atilia Thomas**, REU Student (ICompBio), UTC	*UTC*
Jan 2024 - May 2024	**Quinten Nevenhoven**, MSDS Student, UTC	*UTC*
Sep 2023 - May 2024	**Ja'Lisa Little**, MSDS Student, UTC	*UTC*
Nov 2023 - Apr 2024	**Joseph Ballew**, Master's Student, UTC	*UTC*
Jan 2024 - Apr 2024	**Miscily Bravobussey**, Undergraduate Honor's College Student, UTC	*UTC*

Service and Administrative

Service and Outreach

December 2024	**National Science Foundation**, Servered as reviewer
June 2024	**National Science Foundation**, Servered as reviewer
March 22-24	**MCBIOS 2024, Emory University, GA**, Organised the technical session ”Application of AI in Infectious Diseases/Biological Data Sets”
Summer 2015	**Science Fair, Garfield County, CO**, Volunteer Judge
2016-2018	**Graduate Recruitment Committee (GRC)**, Students' Representative